

MedNexus Game Mode Engine

Comprehensive Architecture & Blueprint

1. The Global Lobby Architecture

The game modes are organized into an intent-driven routing system that splits the user journey based on whether they want an isolated training session or a group experience.

- **The Hero Split:** The entry screen presents a clean two-card navigation system: [ Solo Training] and [ Multiplayer].
- **Solo Routing:** Opens the upgraded solo roster equipped with progression rewards and power-ups.
- **Multiplayer Routing:** Opens the competitive game roster and includes a permanently docked "Join via 6-Digit PIN" input field at the top of the interface.

2. The Upgraded Solo Experience & Mechanics

To prevent solo play from feeling like a raw quiz, the engine incorporates meta-progression, scaling score modifiers, and functional gameplay items.

The Modes

- **Rapid Fire:** A high-speed mode featuring a strict 30-second clock per question and a 3-lives system.
- **Sudden Death:** An endurance mode where a single incorrect answer triggers an immediate game over.
- **Time Attack:** Players start with a 90-second global countdown bank. Correct answers add 3 seconds; incorrect answers instantly drain 5 seconds.
- **Streak Master:** A self-paced mode with no active timers where players build a consecutive correct-answer streak and must choose when to manually "cash out" to secure their rewards.

Engagement Mechanics

- **Dynamic Streak Multipliers:** Building a consecutive correct-answer streak scales the player's performance score (e.g., x1.5 up to x3.0). At a 5x streak, visual indicators (like the timer bar) transition into a high-alert color state, increasing base score rewards per question.
- **In-Match Lifelines (Consumables):** Users can earn or equip single-use lifeline tokens to navigate high-yield clinical questions:
 - *Consult Attending (50/50):* Instantly disables two incorrect multiple-choice options.
 - *Stat Labs (Time Freeze):* Pauses the active question timer for 10 seconds, allowing uninterrupted reading of lengthy clinical vignettes.

3. The Multiplayer Roster

Multiplayer modes are architected around strict synchronization rules to support diverse group sizes and dynamics.

- **Multiplayer Clash (Free-for-All):** Strictly hard-capped at a maximum of 5 players. Players race against the same questions simultaneously, and the full leaderboard is rendered for all participants during a mandatory 5-second pause between questions.
- **Cohort Review (Kahoot Style):** A large-scale lecture-hall mode with no player cap. It utilizes a split-screen layout: the host projects the full clinical vignette and a Top 10 leaderboard on a main display, while individual student devices act purely as input buzzers displaying large, color-coded answer choices (A, B, C, D).
- **Wager Wars:** Players enter the room with a starting balance of virtual currency. The clinical vignette is revealed, but the four diagnostic options remain hidden. Players must place a point wager based purely on their confidence in the text before the options drop. Players who lose all points enter a view-only "Spectator Mode" for the remainder of the match.

4. The Virtual Economy (Nexus Points / Clinical Credits)

A persistent ecosystem rewarding accuracy and engagement, providing utility across both solo and multiplayer environments.

The Earning Engine

- **Base Payouts:** Completing any solo or multiplayer game awards a flat-rate participation payout to ensure continuous progression.
- **Performance Bonuses:** High streaks, podium finishes in Multiplayer Clash, and perfect rounds award scaling currency bonuses.
- **Daily Bounties:** Three rotating objectives are pinned to the dashboard daily (e.g., "*Solve 5 Respiratory questions in Time Attack*"), paying out lump-sum currency rewards upon completion.

The Spending Sinks

- **The Supply Closet (Consumables):** Before launching a high-stakes solo run (like Sudden Death), users can spend accumulated currency to buy and equip their *Consult Attending* or *Stat Labs* lifelines.
- **Premium Clinical Simulations ("The Vault"):** A locked tier of complex, multi-step case studies. Users spend a large sum of currency to permanently unlock deep-dive virtual patient simulations that test case management over time.
- **The Cosmetic Store:** Custom text titles (e.g., *Chief Resident*), animated avatar frames, and leaderboard visual highlights that allow top-tier students to customize their presence during multiplayer match pauses.
- **Wager Wars Buy-In:** Serves as the high-roller room where players must stake their actual wallet balance to compete.

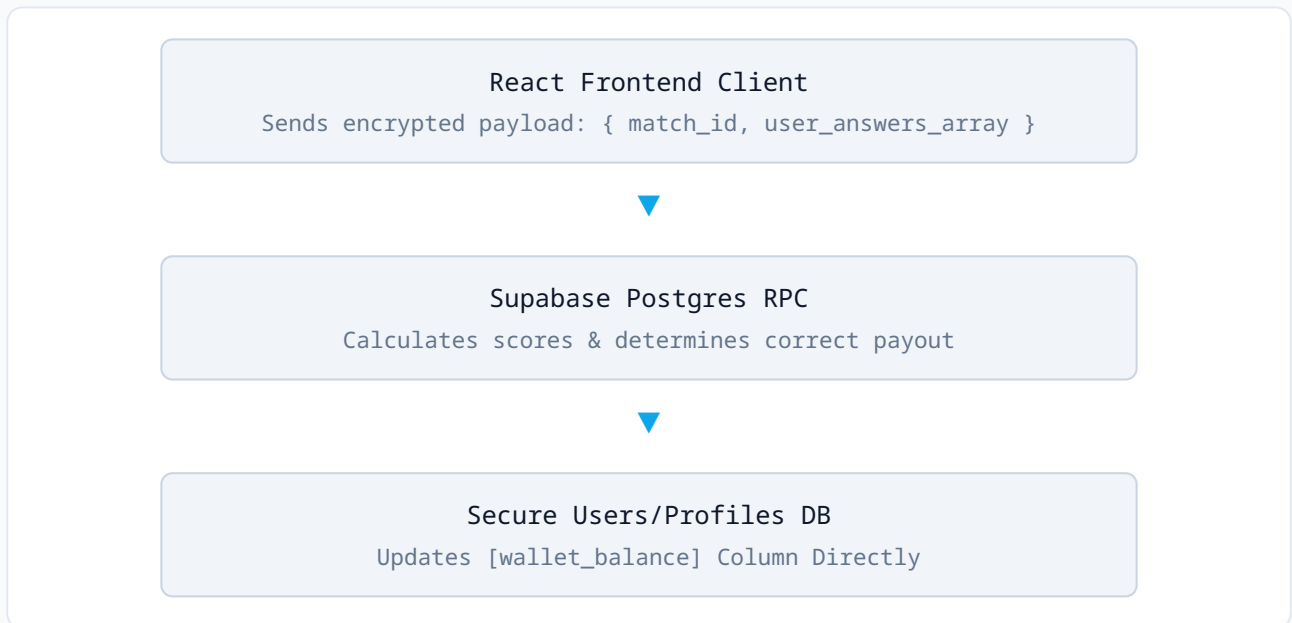
5. Technical Architecture & Eliminating Hindrances

To prevent network desyncs, page-reload state wipes, and hosting bottlenecks, the React and Supabase real-time engines use the following structural rules:

- **Resolving the "Wrong When Correct" Sync Bug:** Completely detach answer verification from local array indexing. When a user selects an option, the client payload sent to the backend database verification function must explicitly include the unique `question_id`. The logic evaluates the equation strictly on the specific question ID matches, ensuring accuracy regardless of network lag.
- **Removing the Host-as-Player Bottleneck:** The match becomes entirely self-driven. The Supabase Realtime engine monitors the active room's Answers table. The exact millisecond the last connected player submits their choice (or when the countdown timer hits zero), the database automatically initiates a 3-second leaderboard flash payload, then automatically pushes the next question index out to all synchronized clients.
- **Mitigating Background Tab Timer Desyncs:** Do not calculate time using incremental ticks. The moment a question renders, record an absolute expiration timestamp: `Expiration Timestamp = Date.now() + 30000`. The UI timer then continuously calculates remaining time by running `Expiration Timestamp - Date.now()`. If a user leaves the tab and returns, the timer will have accurately advanced or expired in real time.
- **Graceful Forfeits & Host Migration:** A prominent [ Leave Match] button is added to the active HUD. Clicking it updates that user's status row in Supabase to `disconnected`. The room's auto-advance engine immediately drops them from the active counts, keeping the room alive. If the host leaves, a Postgres database trigger instantly promotes the oldest remaining active connection to room host.
- **Page Refresh State Recovery:** Cache the current `match_id`, `user room_pin`, and active `question_id` inside the browser's persistent `sessionStorage`. On application mount, a global `useEffect` check scans for an ongoing match session; if one is found, it bypasses the lobby entirely and forces the game board to re-initialize and sync back into the live room state.
- **Post-Match Answer Review Side-Drawer:** The final scoreboard screen houses a prominent [ Review Vignettes] button. When clicked, a wide side-drawer surfaces a scrollable list of every question from that specific match session. It renders the complete clinical vignette, a red highlight over the user's wrong choice (if applicable), a green highlight over the correct option, and the complete text explanation (EXP) pulled from your database.

6. Secure Database Architecture (The Wallet Ledger)

To secure the in-game currency against frontend manipulation (such as users executing client-side scripts to artificially inflate their balances), all transaction logic is centralized within protected backend systems.



- **The Data Constraints:** The user's wallet balance exists as a protected integer column on a server-side profiles table.
- **The Execution:** When a game concludes, the client transmits only the raw transaction details (the `match_id` and the user's `answers_array`). It never transmits a point value. A database function calculates the accuracy, matches it against the room type, and securely writes the updated point total directly to the server ledger.